STEEL REINFORCING BARS AND WIRE FABRIC

MES-ST-SPC-0006

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for the supply of steel reinforcing bars and wire fabric for Ma'aden Projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum standard for materials, drawings, fabrication and supply of steel reinforcing bars and wire fabric.

1.3 Conflicts and Deviations

Any deviation from any of the requirements stipulated in this document shall go thru a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In

general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All buildings, equipment and systems shall conform to the latest adopted revision of the Codes and standards in the following paragraphs. In case of conflict in general the most stringent requirements shall govern.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

SBC 304	Structural - Concrete Structures
SBC 801	Fire protection

Applicable local codes for the Project/Site

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 *International Codes and Standards*

3.2.2.1. American Concrete Institute (ACI)

ACI PRC-315	Guide to Presenting Reinforcing Steel Design Details
ACI CODE-318	Building Code Requirements for Structural Concrete

3.2.2.2. American Society for Testing and Materials

ASTM A615/A615M	Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A767/A767M	Standard Specification for zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement
ASTM A775M/A775M	Standard Specification for Epoxy-Coated Reinforcing Bars
ASTM A884/A884M	Standard Specification for Epoxy-Coated Steel Wire and Welded Wire Reinforcement
ASTM A934/A934M	Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars
ASTM A1055/A1055M	Standard Specification for zinc and Epoxy Dual-Coated Steel Reinforcing Bars
ASTM A1064/A1064M	Standard Specification for Carbon- Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM D7957/D7957M	Standard Specification for Solid Round Glass Fiber Reinforced Polymer Bars for Concrete

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0004	Pre-Mixed Concrete
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS

Materials permitted for use as reinforcement are specified. Other metal elements, such as inserts, anchor bolts, or plain bars for dowels at isolation or contraction joints, are not normally considered to be reinforcement under the provisions of SBC 304.

All reinforcing bars and wire fabric shall be fabricated from the steel grade specified on the drawings and shall be free from rust, scaling and pitting and from coatings which would reduce or destroy concrete bond.

All reinforcing bars shall comply with ASTM requirement noted in SBC 304, Steel Reinforcement. Reinforcing bars shall be deformed bar conforming to ASTM A615/A615M grade 60 with minimum yield strength 420 MPa and minimum tensile strength 550 MPa, unless noted otherwise on the drawings.

Tie wire shall be black annealed wire, minimum 1.6 mm diameter. Epoxy coated wire shall be used wherever epoxy coated bars are used.

Galvanized reinforcing bars (A 767M), epoxy-coated reinforcing bars (A 775M) and epoxy-coated prefabricated reinforcing bars (A 934M) and zinc and epoxy dual-coated steel reinforcing bars (A1055M) are used, especially for conditions where corrosion resistance of reinforcement is of particular concern.

The Type of Reinforcement shall be as per project specification or Contract drawings.

Any conflicts between this document and other applicable Codes, Specifications, Standard Drawings, Ma'aden Standards or any other Industry standards, shall be brought to the attention of Employer Representative by the Contractor for resolution.

Welded wire fabric shall normally be plain wire fabric and conform to ASTM A1064/A1064M. Welded plain wire fabric should be made of wire conforming to "Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete"(ASTM A1064/1064M) with minimum yield strength 420 MPa.

Concrete Structures that are cathodic protected shall be reinforced with uncoated steel rebars conforming to ASTM A615 grade 60. Binding wires shall also be uncoated.

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5. EXECUTION

5.1 Quality of Execution

All reinforcement bars shall be accurately placed, properly supported and firmly attached to prevent displacement during works and the pouring of concrete.

An original copy of the mill test reports, which include the physical and chemical analysis of the steel supplied, shall be submitted to the Employer, at least two weeks prior to starting the reinforcing works.

5.2 Shaping, Bending and Installation of Reinforcing Steel

It shall be the Contractor's responsibility to provide a complete bar bending schedule showing at least the bar mark numbers, bar shape, material type, length before bending, number of bars required and total weight for each group of bars. The bar bending schedule shall ensure adequate concrete cover to all reinforcing bars as shown on the drawings or as specified in SBC 304 and SBC 801. Reinforcing shall be detailed and bent in accordance with ACI PRC-315. Heating of bars to facilitate bending shall not be permitted. Bending or straightening of bars after their embedment in concrete shall not be permitted.

The Contractor will be provided with copies of the Contract drawings. The Contractor shall add bar mark numbers together with bar schedule numbers to these drawings and return them together with the bar schedules to the Employer.

Except as otherwise specified or shown on the drawings, all fabrication of hooks and bends, including bends at a cranked portion of a bar, shall be bent cold and shall conform to the minimum requirements of ACI PRC-315. Tolerance on overall dimensions of any cut or bent reinforcing bar or fitment shall be in accordance with ACI PRC-315.

All splice, tying wire, chairs and riser bars shall be accordance with ACI CODE-318, and standard specification MD-101-SMEM-EG-ST-SPC-0006, Concrete Work, Section 4.2.

Welding of reinforcement bars shall NOT be considered as a recommended practice in accordance with SBC 304.

In the event conduits, piping, inserts, sleeves or other items interfere with reinforcement placing, the Contractor shall immediately notify Employer and obtain approval for adjustments. The Contractor shall not cut or move bars to clear obstructions without Employer's approval.

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Reinforcing bars shall be spliced in accordance with design drawings only. Unless otherwise approved by Employer, welded splices shall not be permitted.

5.3 Identification

Groups of identical bars shall be securely tied together for shipment and tagged according to mark numbers and bar schedule number. Tags shall consist of stamped aluminum or zinc strips.

5.4 Handling and Storage

All necessary supervision, chains, timber dunnage, tie-down equipment, flags, permits and the like shall be provided to safely and carefully complete transportation of reinforcement steel. Reinforcement steel shall be stacked clear of the ground and be self draining. Wooden supports in contact with finished items during transport and storage shall be covered with a soft material such as rubber to minimize damage to the coatings. Projections from steelwork such as lugs and splice plates which may suffer damage during transport are required to be reinforced with a timber packing piece.

5.5 Deliveries

The Contractor shall be responsible for the scheduling of deliveries to suit the programmed work requirements.

The Contractor shall be responsible for the safe undamaged delivery of all reinforcing steel to the designated area on site and for the safe storage on site until required for use.

5.6 Inspection

Inspection of the carbon-steel reinforcing bars shall be agreed upon between the purchaser and the manufacturer as part of the purchase order or contract.

The Contractor shall afford the Employer all reasonable facilities to inspect the material and fabrication prior to delivery to ensure compliance with this specification.

Any such inspection by the Employer shall not relieve the Contractor of any responsibility as required by the Contract.

The inspection procedure shall comply with MD-101-SMPM-PM-QA-PEI-0001, Contractor Quality Requirements.

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